

Star Dataset: Report

Machine Learning Star Class Prediction & Analysis



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Research Question

How do the intrinsic physical properties of a star—specifically its temperature, luminosity, radius, and absolute magnitude—determine its stellar classification, and to what extent can these variables be used to accurately predict star types within the framework of celestial dynamics?

Star Dataset (DF)

- Absolute Temperature (in K)
- Relative Luminosity (L/Lo)
- Relative Radius (R/Ro)
- Absolute Magnitude (Mv)
- Star Color (White, Red, Blue, Yellow, yellow-orange, etc.)
- Spectral Class (O, B, A, F, G, K, M)
- Star Type (Red Dwarf, Brown Dwarf, White Dwarf, Main Sequence, SuperGiants, HyperGiants)

Dataset Cleaning

This dataset did not contain any missing values, however, not all the columns were in numerical values so I had to complete the encoding.

df.dtypes		df.isna().sum()	
Temperature (K)	int64	Temperature (K)	0
Luminosity(L/Lo)	float64	Luminosity(L/Lo)	0
Radius(R/Ro)	float64	Radius(R/Ro)	0
Absolute magnitude(Mv)	float64	Absolute magnitude(Mv)	0
Star type	int64	Star type	0
Star color	object	Star color	0
Spectral Class	object	Spectral Class	0
dtype: object		dtype: int64	

Encoding:

The columns I had to encode were Star Color and Spectral Class. I began by determining the unique values for each column and then mapping each non-numerical value to a number. Then I applied the map to the original column, defining it as a new encoded column. To simplify the colors I mapped them to only those found on roygiv.

```

import numpy as np
def str_cleanup(Str):
    Str = Str.capitalize() # Capitalize the first letter
    Str = Str.strip()      # Remove leading and trailing whitespaces
    Str = Str.replace('-', ' ') # Replace hyphens with spaces
    return Str

# Assuming X_train is your DataFrame and 'Star color' is the column containing color labels
df['Star Color Map'] = np.vectorize(str_cleanup)(df['Star color'])

```

```
[ ] df['Star Color Map'].unique()
```

```

array(['Red', 'Blue white', 'White', 'Yellowish white',
      'Pale yellow orange', 'Blue', 'Whitish', 'Yellow white', 'Orange',
      'White yellow', 'Yellowish', 'Orange red'], dtype=object)

```

```

[ ] color_map = {
    'Red':0, 'Blue white':1, 'White':2, 'Yellowish white':3,
    'Pale yellow orange':4, 'Blue':5, 'Whitish':6, 'Yellow white':7, 'Orange':8,
    'White yellow':9, 'Yellowish':10, 'Orange red':11
}

```

```
[ ] df['Spectral Class'].unique()
```

```

array(['M', 'B', 'A', 'F', 'O', 'K', 'G'], dtype=object)

```

```
[ ] Spectral_Class_Map = { 'M': 0, 'B': 1, 'A':2, 'F':3, 'O':4, 'K':5, 'G':6}
```

```
[ ] df['Spectral Class Map'] = df['Spectral Class'].map(Spectral_Class_Map)
```

```

[ ] color_map_plot = { 'Red':'red', 'Blue white':'white', 'White':'white', 'Yellowish white':'white',
    'Pale yellow orange':'orange', 'Blue':'blue', 'Whitish':'white', 'Yellow white':'white', 'Orange':'orange',
    'White yellow':'yellow', 'Yellowish':'yellow', 'Orange red':'red' }

```

```
[ ] colors = df['Star color'].map(color_map_plot)
```

Machine Learning

Although this dataset already contained the Star type predictions I wanted to run my own model and compare my result to the listed value. For my machine learning program, I used the xgboost classification model. For my features, I used 'Temperature (K)', 'Luminosity(L/Lo)', 'Radius(R/Ro)', 'Absolute magnitude(Mv)', and 'Star Color Code'. After fitting the model I compared my model's predictions to the listed values of the dataset and had a 100% accuracy.

df.columns

```
Index(['Temperature (K)', 'Luminosity(L/Lo)', 'Radius(R/Ro)',  
      'Absolute magnitude(Mv)', 'Star type', 'Star color', 'Spectral Class',  
      'Star Color Map', 'Star Color Code', 'Spectral Class Map'],  
      dtype='object')
```

```
[20] import xgboost as xgb
```

```
[21] features = ['Temperature (K)', 'Luminosity(L/Lo)', 'Radius(R/Ro)', 'Absolute magnitude(Mv)',  
              'Star Color Code']
```

```
[22] X = df[features]  
     y = df['Star type']
```

```
# Initialize XGBoost model  
model = xgb.XGBClassifier()  
  
# Fit the model to the entire dataset  
model.fit(X, y)
```

```
XGBClassifier  
XGBClassifier(base_score=None, booster=None, callbacks=None,  
              colsample_bylevel=None, colsample_bynode=None,  
              colsample_bytree=None, device=None, early_stopping_rounds=None,  
              enable_categorical=False, eval_metric=None, feature_types=None,  
              gamma=None, grow_policy=None, importance_type=None,  
              interaction_constraints=None, learning_rate=None, max_bin=None,  
              max_cat_threshold=None, max_cat_to_onehot=None,  
              max_delta_step=None, max_depth=None, max_leaves=None,  
              min_child_weight=None, missing=nan, monotone_constraints=None,  
              multi_strategy=None, n_estimators=None, n_jobs=None,  
              num_parallel_tree=None, objective='multi:softprob', ...)
```

```
[24] df['My Prediction'] = model.predict(X)
```

```
from sklearn.metrics import accuracy_score  
  
# Calculate accuracy using scikit-learn  
accuracy = accuracy_score(df['Star type'], df['My Prediction'])  
print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

Data Analysis & Visualization

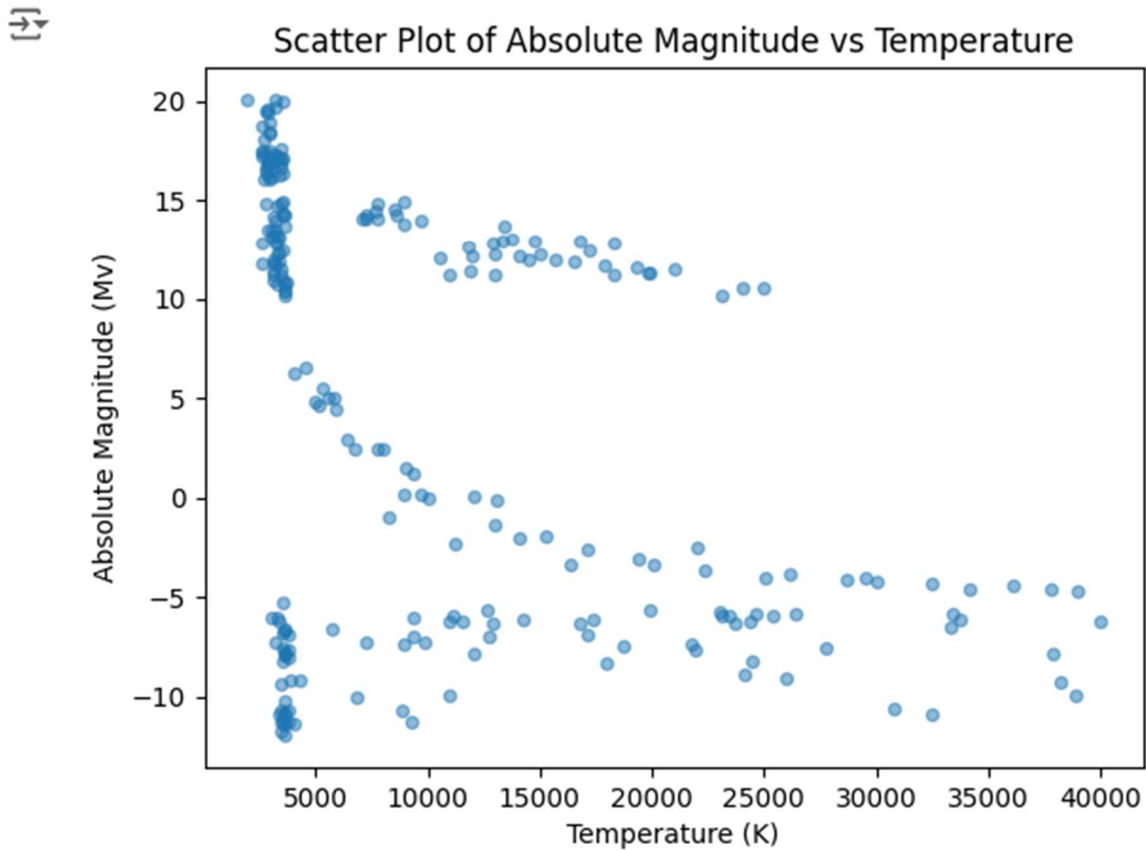
Google Colab:

The goal of my data analysis was to find the defining qualities of various star groups.

```
# Create a scatter plot
df.plot.scatter(x='Temperature (K)', y='Absolute magnitude(Mv)', alpha=0.5)

# Add labels and title
plt.title('Scatter Plot of Absolute Magnitude vs Temperature')
plt.xlabel('Temperature (K)')
plt.ylabel('Absolute Magnitude (Mv)')

# Show the plot
plt.show()
```




```

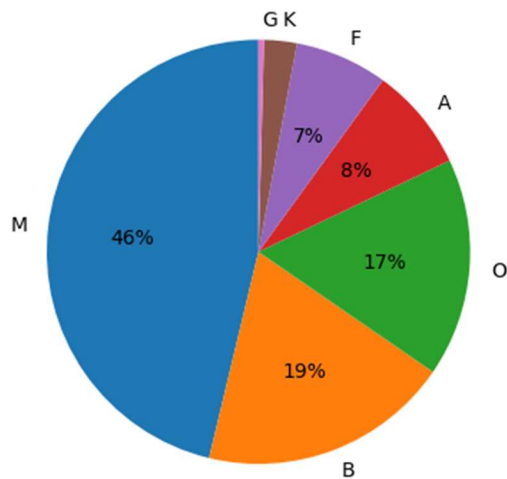
# Counting the frequency of each Spectral Class
class_distribution = df['Spectral Class'].value_counts()

# Plotting pie chart
class_distribution.plot(kind='pie', autopct=lambda p: '{:.0f}%'.format(p) if p > 5 else '', startangle=90)
plt.title('Distribution of Spectral Classes')
plt.ylabel('')
plt.show()

```



Distribution of Spectral Classes



```

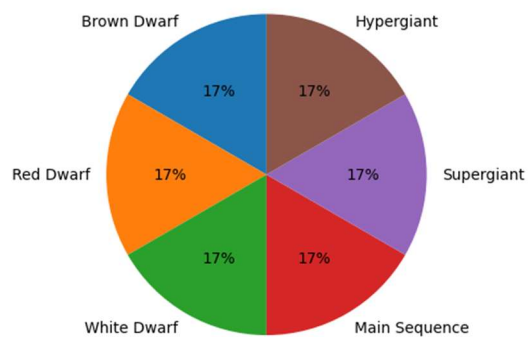
# Counting the frequency of each MP Star Type String
class_distribution = df['MP Star Type String'].value_counts()

# Plotting pie chart
class_distribution.plot(kind='pie', autopct=lambda p: '{:.0f}%'.format(p) if p > 5 else '', startangle=90)
plt.title('Distribution of MP Star Type Stringes')
plt.ylabel('')
plt.show()

```



Distribution of MP Star Type Stringes



```
# Scatter plot: Absolute Magnitude vs Temperature for all stars
plt.figure(figsize=(8, 6))
plt.scatter(temperature, absolute_magnitude, alpha=0.5, color='blue')
plt.title('Absolute Magnitude vs Temperature for All Stars')
plt.xlabel('Temperature (K)')
plt.ylabel('Absolute Magnitude (Mv)')
plt.grid(True)
plt.show()
```

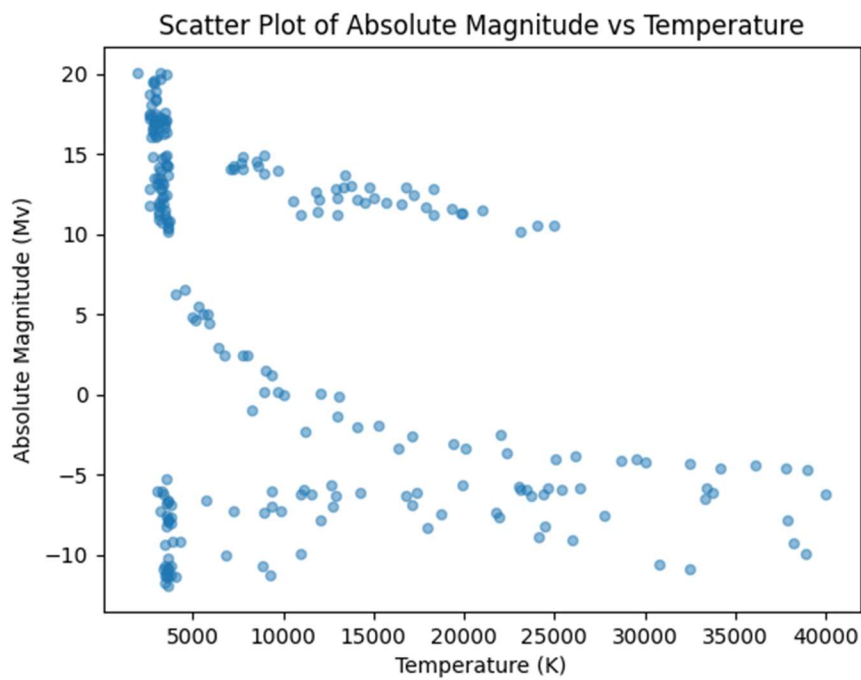
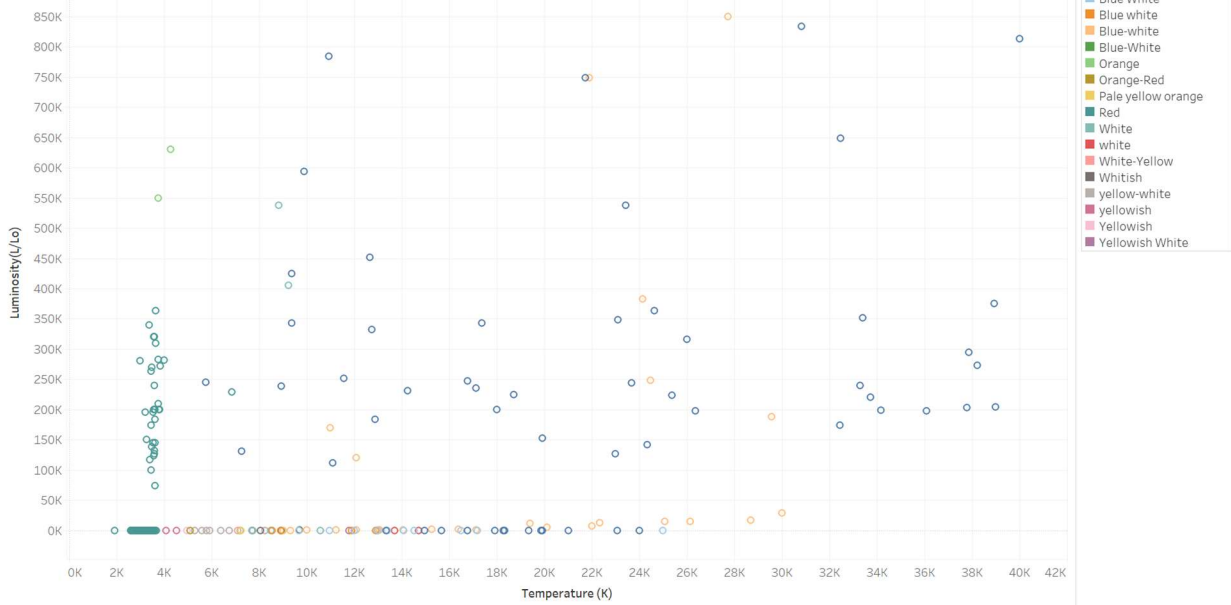


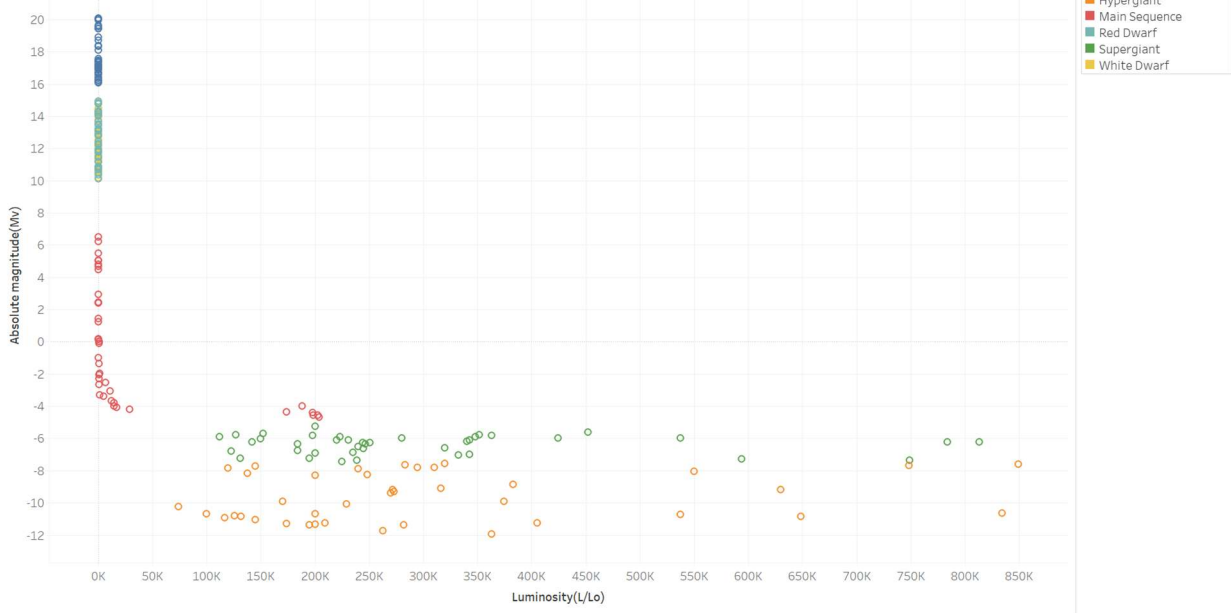
Tableau:

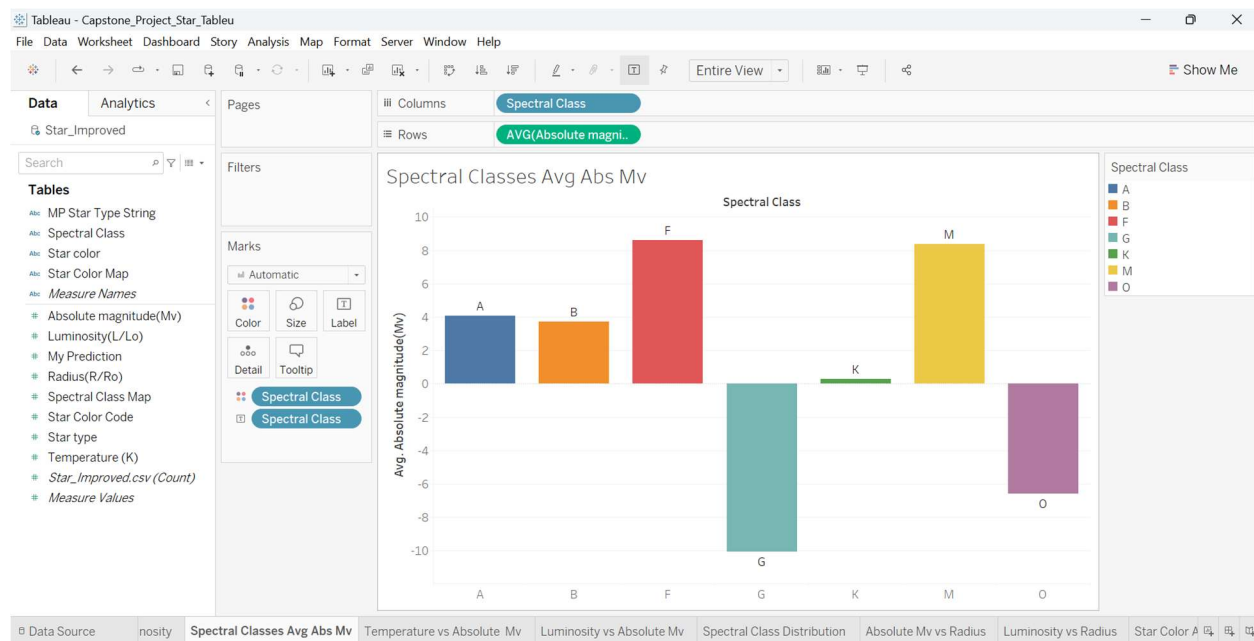
An even more efficient way to make quality data visualizations is using the Tableau software which allows for better customization. Here are some visualizations I made.

Temperature vs Luminosity

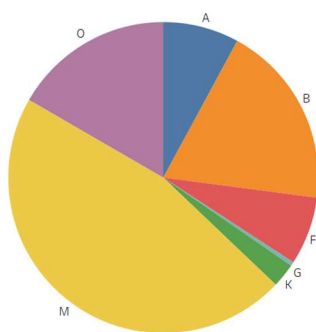


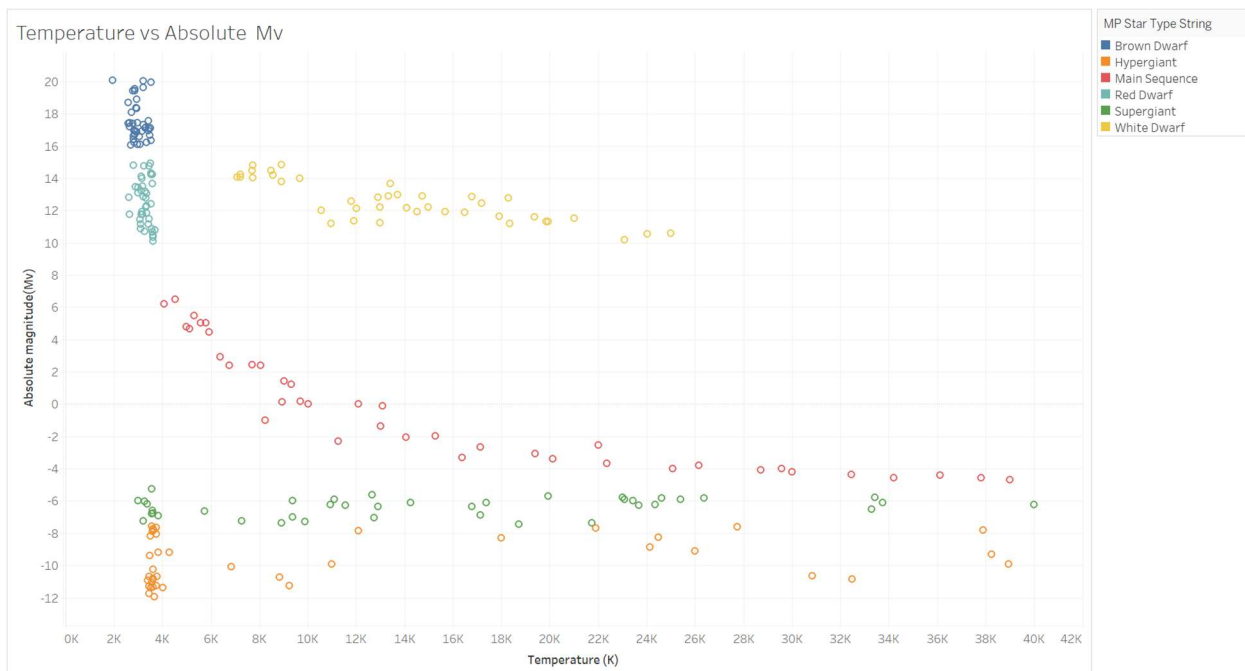
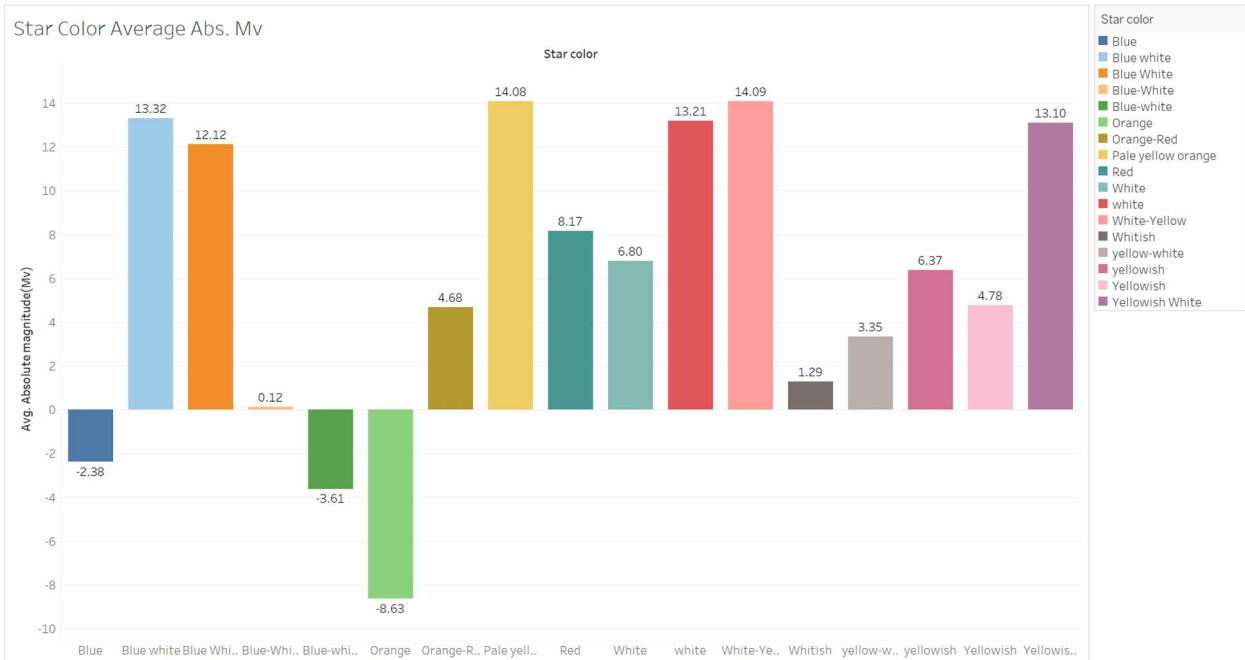
Luminosity vs Absolute Mv





Spectral Class Distribution





Concluding Thoughts

In this data science journey, I delved into a comprehensive stars dataset to uncover the scientific qualities underpinning star types. My objective was to decipher the intrinsic characteristics that determine the classification of stars, contributing to a deeper understanding of celestial dynamics.

Through analysis and visualization, I explored key attributes such as temperature, luminosity, radius, and absolute magnitude. Leveraging advanced machine learning models, I achieved 100% accuracy in predicting star types, highlighting the predictability and structure inherent in the dataset.

Visualization techniques revealed compelling relationships between luminosity and radius, as well as absolute magnitude and temperature. These insights shed light on the physical properties and evolutionary stages of stars, clarifying the scientific qualities that define their classifications.

Furthermore, the distribution of star types, colors, and spectral classes unveiled intriguing patterns indicative of underlying stellar phenomena. By unraveling these complexities, I have contributed to a deeper understanding of celestial dynamics and the scientific principles governing star classifications.

This study underscores the power of data science in unraveling the mysteries of the universe and advancing our scientific understanding of stellar phenomena.

References

1. <https://www.kaggle.com/datasets/deepu1109/star-dataset>